R.E.A.P : Realtime Edge Agentic Physio-Buddy

B.Tech (IT) Major Project — Status Update

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Problem Statement

- Form degrades long before fatigue is felt.
- Bad form mid-set $\rightarrow$ joint / tendon injury (older adults at highest risk).
- No real-time self-correction without a coach.
- Need: instant, eyes-free audio feedback every rep.

Objectives

- O1 — Instant feedback: sub-second audio cues mid-rep.
- O2 — Injury prevention: catch velocity-loss + deviation, stop before failure.
- O3 — Accessibility: phone-grade kinematic tracking at home.

System Architecture

- Camera → Hybrid Pose (MediaPipe + YOLO-Pose)
- → Predictive Kinematics (OneEuro now, N-euro later)
- → Action Recognition (ST-GCN) → Fatigue Model → Local TTS Audio
- Constraint: 100% on-device, zero cloud, audio-first.

Work Done (Sept milestone)

- Project scaffold: uv + pyproject, configs, modular src/reap/.
- Hybrid pose core: MP every frame + YOLO fallback / validator, COCO-17 fusion.
- Benchmark app: live MP vs YOLO vs Hybrid + FPS / latency HUD + CSV log.
- Kinematics + smoothing: joint angles, OneEuro filter, N-euro stub.
- Verified: 6/6 tests pass; MP ~32 ms/frame; hybrid fallback works.

Pending / Next Work

- Oct–Nov: core framework + exercise plug-in packs + edge tuning.
- Dec: local TTS + audio cue engine.
- Jan–Feb: predictive fatigue (velocity-loss) modeling.
- Mar: real-world testing with older adults. Apr: N-euro / ST-GCN iteration.

Thank You

Demo: uv run reap-bench --source 0 --mode hybrid

Questions welcome.